



NORTH SYDNEY BOYS HIGH SCHOOL

Mathematics Extension 1

2024 Assessment Task 3

General Instructions:

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- For questions in Section B, show all relevant mathematical reasoning and/or calculations
- Use a new booklet for each question
- Write your student number and tick teacher name on **everything**

Teacher

(Please tick)

- ☐ Mr Berry
- ☐ Ms Cai
- ☐ Mr Ireland
- ☐ Ms Lee
- ☐ Ms Moss
- ☐ Mr Umakanthan
- ☐ Dr Vranešević

Student Number:

4								
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Marker Use Only:

Question	MC	11	12	13	14	Total	%
Mark	$\overline{10}$	$\overline{15}$	$\overline{15}$	$\overline{15}$	$\overline{15}$	$\overline{70}$	

Section A – Multiple Choice (10 marks)

1. Given $\underline{u} = 2\underline{i} - 6\underline{j}$ and $\underline{v} = -\underline{i} + 5\underline{j}$, $\underline{u} - \underline{v}$ is equivalent to:

A. $\underline{i} - \underline{j}$

B. $3\underline{i} - \underline{j}$

C. $\underline{i} - 11\underline{j}$

D. $3\underline{i} - 11\underline{j}$

2. Let α, β and γ be the roots of $x^3 + px^2 + q = 0$.

Express $\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\alpha\gamma}$ in terms of p and q

A. pq

B. $-pq$

C. $-\frac{p}{q}$

D. $\frac{p}{q}$

3. Harry projects an arrow at an angle of 60° to the horizontal with an initial velocity of 30 ms^{-1} . What is the horizontal speed of the arrow?

A. 60 ms^{-1}

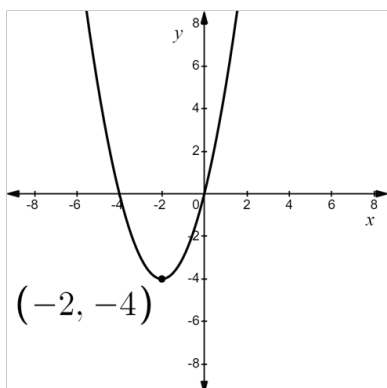
B. $\frac{30}{\sqrt{3}} \text{ ms}^{-1}$

C. $30\sqrt{3} \text{ ms}^{-1}$

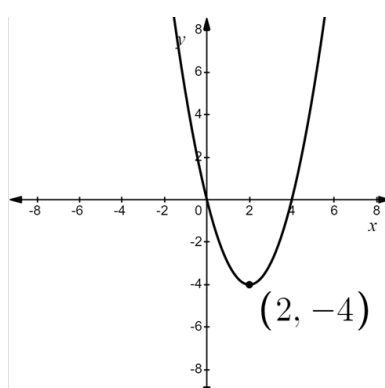
D. 15 ms^{-1}

4. Given $f(x) = x^2 - 4$ and $g(x) = |-x - 2|$, which graph represents $f(g(x))$?

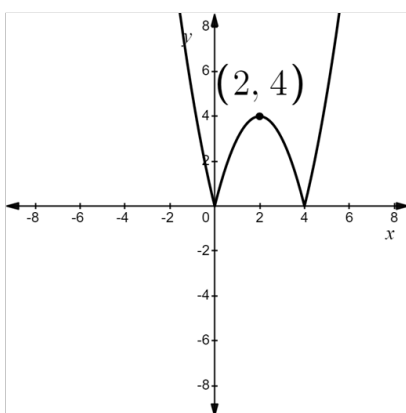
A.



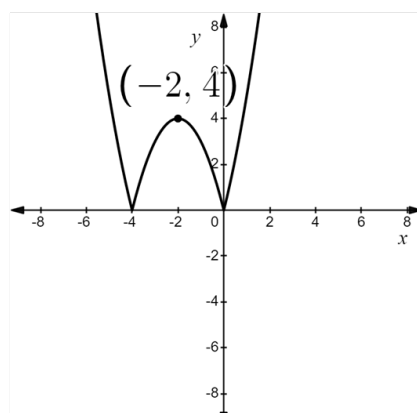
B.



C.



D.



5. If $\cos x = -\frac{4}{5}$ and $\frac{\pi}{2} \leq x \leq \pi$ then $\tan 2x$ is equal to:

A. $-\frac{24}{7}$

B. $\frac{24}{7}$

C. $\frac{12}{7}$

D. $-\frac{12}{7}$

6. Given $\tilde{u} = 2\tilde{i} + 3\tilde{j}$ and $\tilde{v} = -2\tilde{i} + 4\tilde{j}$, $proj_{\tilde{u}}\tilde{v}$ is:

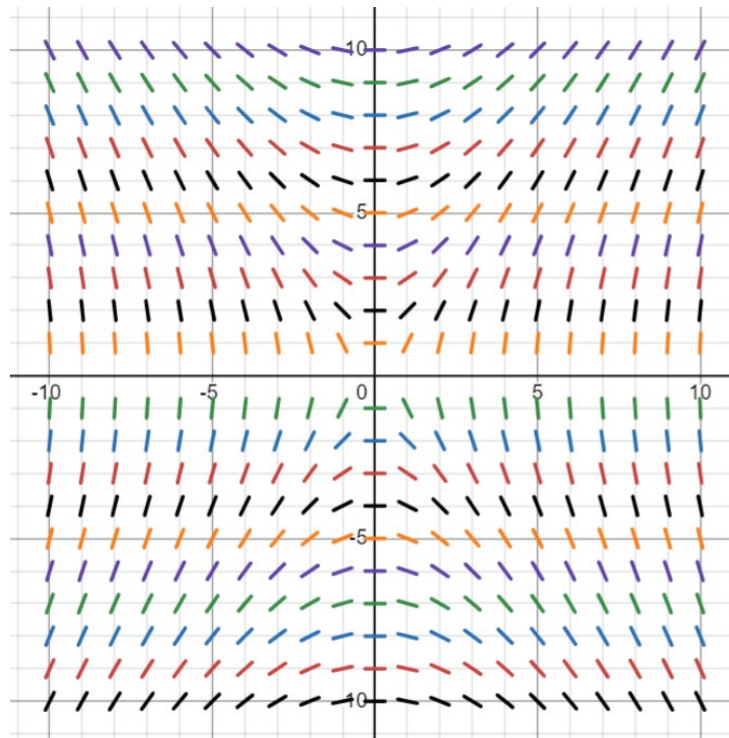
A. $-\frac{16}{13}\tilde{i} + \frac{32}{13}\tilde{j}$

B. 8

C. $\frac{8}{13}$

D. $\frac{16}{13}\tilde{i} + \frac{24}{13}\tilde{j}$

7. Which of the following differential equations could be represented by the slope field drawn below?



- A. $\frac{dy}{dx} = \frac{2x}{y}$ B. $\frac{dy}{dx} = \frac{2y}{x}$
- C. $\frac{dy}{dx} = \frac{x^2}{y}$ D. $\frac{dy}{dx} = \frac{x}{y^2}$

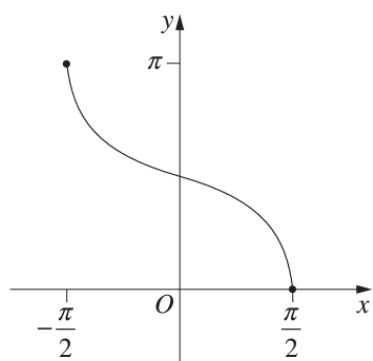
8. A curve C has parametric equations $x = \cos^2 t$ and $y = 4 \sin^2 t$ for $t \in R$.

What is the Cartesian equation of C ?

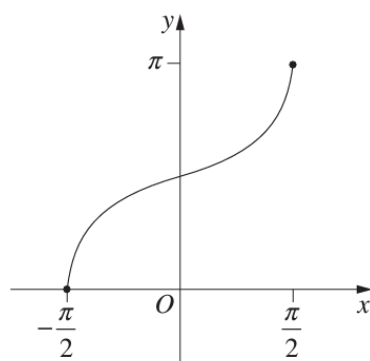
- A. $y = 1 - x$ for $0 \leq x \leq 1$ B. $y = 4 - 4x$ for $x \in R$
- C. $y = 1 - x$ for $x \in R$ D. $y = 4 - 4x$ for $0 \leq x \leq 1$

9. Which graph best represents $y = \cos^{-1}(-\sin x)$, for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

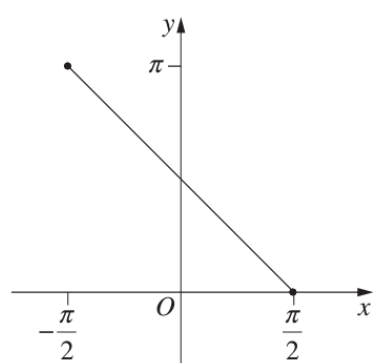
A.



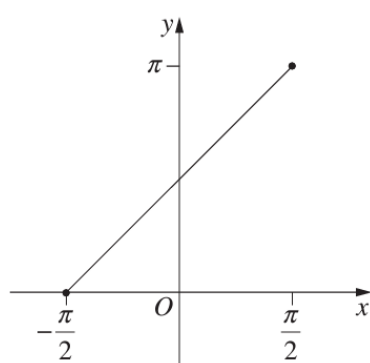
B.



C.

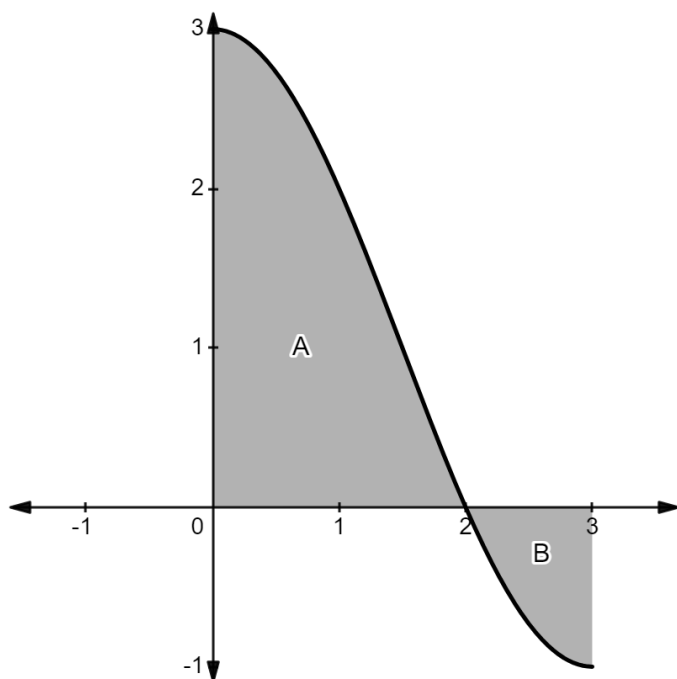


D.



10.

The graph of $y = 2 \cos \frac{\pi x}{3} + 1$ is shown



Region A has an Area of $\frac{3\sqrt{3}}{\pi} + 2$ units

Region B has an Area of $\frac{3\sqrt{3}}{\pi} - 1$ units

Using this information, evaluate $\int_{-1}^3 \frac{3}{\pi} \cos^{-1} \frac{x-1}{2} dx$

A. 3

B. 6

C. $\frac{6\sqrt{3}}{\pi} + 1$

D. $11 - \frac{6\sqrt{3}}{\pi}$

Section B – Questions 11-15 (60 marks total)

Question 11 (15 marks)

(START A NEW BOOKLET)

- a) Solve for x : 2

$$\frac{6}{x-2} \geq 3$$

- b) Find the exact value of: 2

$$\sin\left(2 \cos^{-1}\left(\frac{2}{3}\right)\right)$$

- c) Find, in simplest terms, the coefficient of the x^6 term in the expansion of: 2

$$\left(2x^2 - \frac{1}{3x}\right)^9$$

- d) How many numbers greater than 8000 can be formed from the digits 1, 2, 4, 6, 9 if no digit is repeated? 2

- e) Use $t = \tan \frac{\theta}{2}$ to solve 2

$$\sin \theta + \cos \theta = -\frac{1}{4} \text{ for } -\pi \leq \theta < \pi$$

- f) Given $y = f(x)$ where $f(x) = (x-1)^2 - 4$. Sketch the following curves on separate graphs, each at least one-third of a page in size:

(i) $y = -f(2-3x)$ 3

(ii) $y = |f(|x|)|$ 2

Question 12 (15 marks)

(START A NEW BOOKLET)

- a) Using the substitution $u^2 = x + 1$ where $u > 0$ to find: 3

$$\int_0^3 \frac{x+2}{\sqrt{x+1}} dx$$

- b) Given $f(x) = \sin^{-1} x$ and $g(x) = \cos^{-1} x$

- (i) Sketch $f(x)$ and $g(x)$ on the same set of axes 2

- (ii) Hence, state the value of $f(x) + g(x)$ for $-1 \leq x \leq 1$ and explain the significance of this finding. 1

- c) By methods of induction, prove that $3^{2n+2} - 8n - 9$ is divisible by 64 for all integers $n \geq 1$ 3

- d) Show that if 19 distinct numbers are chosen from the sequence 1, 4, 7, 10, ..., 100 there must be two of them whose sum is 104. 2

- e) Given $4x^4 + 8x^3 + 3x^2 - 2x - 1 = 0$

- (i) Express in the form $(ax^2 + bx)^2 - (cx + d)^2 = 0$, where a, b, c and d are positive integers, and state the values of a, b, c and d . 2

- (ii) Hence solve the equation for x 2

Question 13 (15 marks)

(START A NEW BOOKLET)

a) Given $f(x) = x \cos^{-1} x - \sqrt{1 - x^2}$

(i) Find $f'(x)$ 2

(ii) Hence, evaluate: 2

$$\int_0^1 \cos^{-1} x \, dx$$

- b) Let T be the temperature inside B15 at time t and let A be the constant outside air temperature. Newton's law of cooling states that the rate of change of the temperature T is proportional to $(T - A)$. It can be shown that $T = A + Ce^{kt}$ where C and k are constants satisfies Newton's law of cooling.

The outside air temperature is 15°C and Year 12 come in from lunch and open all the windows. The temperature inside drops from 25°C to 21°C over a period of half an hour.

(i) Find the values of C and k 2

(ii) How much longer would it take the temperature to drop to 16°C ? Give your answer to the nearest minute. 1

- c) A coach is watching a gridiron player from a point O on the sideline. He looks directly at the player without taking his eyes off him. The player starts at position A with position vector $\vec{OA} = 25\vec{i} + 30\vec{j}$. The player runs in a straight line with a constant speed. After 2 seconds, the player is at position B with position vector $\vec{OB} = 19\vec{i} + 33\vec{j}$. The coach watches the player run for 10 seconds in total starting from position A and finishing at position C .

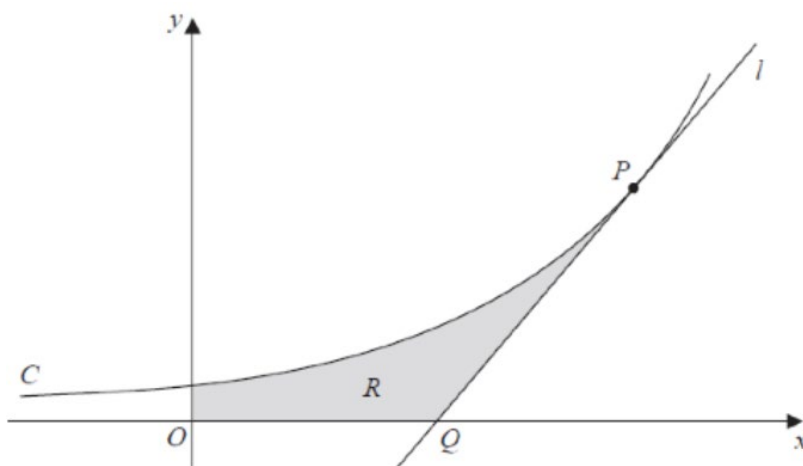
(i) After 10 seconds, what is the position vector $\vec{OC}$ of the player relative to the coach? 2

(ii) Through what angle does the coach turn his head in order to watch them for the full 10 seconds? Answer to the nearest minute. 2

Question 13 (cont.)

- d) The diagram shows a region bound by the curve C with the equation $y = 3^x$ the line l and the x -axis.

The x -co-ordinates of points P and Q are 2 and $(2 - \frac{1}{\ln 3})$ respectively.



A solid is created when the region R is rotated around the x -axis. Find the volume of the solid formed.

Question 14 (15 marks)

(START A NEW BOOKLET)

a) Solve:

$$6 \sin^2 x + 4 \sin x \cos x - 3 \sin x = 0 \quad \text{for } 0 \leq x \leq \pi \quad 3$$

b) Given

$$f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

(i) Show that $y = f(x)$ has no stationary points. 2

(ii) Given that $y = \pm 1$ are horizontal asymptotes, sketch the curve. 2

(iii) For $k > 0$, consider the area enclosed by the curve, the lines $y = 1$, $x = 0$ and $x = k$. Show that this area can be expressed in the form: 2

$$\ln \left(\frac{2e^k}{e^k + e^{-k}} \right)$$

(iv) Hence, justify why for all values of k , the area found in part (iii) is always less than $\ln 2$. 1

c) A particle is projected from a point O with speed 80 ms^{-1} at an angle of elevation α , where $\tan \alpha = \frac{5}{12}$. Two seconds later, a second particle is projected from O and it collides with the first particle one second after leaving O . Let β be the initial angle of projection for the second projectile. Let $g = 10 \text{ ms}^{-2}$.

(i) Show that the displacement of the first particle after t seconds is given by the equation:

$$s(t) = 80t \cos \alpha \mathbf{i} + (80t \sin \alpha - 5t^2) \mathbf{j} \quad 1$$

(ii) Find $\tan \beta$ 3

(iii) Find the initial velocity of the second particle 1

END OF EXAMINATION

Multiple Choice Answers

1. D :

$$\mathbf{u} - \mathbf{v} = (2\mathbf{i} - 6\mathbf{j}) - (-\mathbf{i} + 5\mathbf{j}) = 2\mathbf{i} + \mathbf{i} - 6\mathbf{j} - 5\mathbf{j} = 3\mathbf{i} - 11\mathbf{j}$$

2. D :

$$\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\gamma\alpha} = \frac{\alpha + \beta + \gamma}{\alpha\beta\gamma} = \frac{-p}{-q} = \frac{p}{q}$$

3. D :

$$v \cos 60^\circ = 30 \times \frac{1}{2} = 15 \text{ ms}^{-1}$$

4. A :

$$\text{Note when } x = -2, g(x) = 0, f(x) = -4.$$

5. A :

$$\text{Either by } \sin^2 x + \cos^2 x = 1 \text{ or Pythagoras theorem (adjust for 2nd quadrant, only cos is +ve),}$$
$$\sin x = \frac{3}{5} \text{ and } \tan x = \frac{-3}{4} \Rightarrow \tan 2x = \frac{2 \tan x}{1 - \tan^2 x} = \frac{2 \times \frac{-3}{4}}{1 - \left(\frac{-3}{4}\right)^2} = \frac{-24}{7}$$

6. D :

$$\text{proj}_{\mathbf{u}} \mathbf{v} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{u} \cdot \mathbf{u}} \mathbf{u}$$

$$\mathbf{u} \cdot \mathbf{v} = (2)(-2) + (3)(4) = -4 + 12 = 8$$

$$\text{proj}_{\mathbf{u}} \mathbf{v} = \frac{8}{13} \mathbf{u} = \frac{8}{13} (2\mathbf{i} + 3\mathbf{j}) = \frac{16}{13} \mathbf{i} + \frac{24}{13} \mathbf{j}$$

7. A:

Cannot be B as slope is 0 when $x = 0$ not undefined.

Cannot be C or D because slope is +ve when $x < 0$ and $y < 0$

8. D:

$$\text{Since } \cos^2 t + \sin^2 t = 1 \Rightarrow x + \frac{y}{4} = 1 \Rightarrow y = 4 - 4x$$

9. D:

$$y = \cos^{-1}(-\sin x) = \pi - \cos^{-1}(\sin x) = \pi - \left(\frac{\pi}{2} - x\right) = x + \frac{\pi}{2}$$

10. B:

The integral is equal to the area between the curve and the y axis, which is $A - B + 3$

Section B

Question 11

a) $\frac{6}{x-2} \geq 3$

$\frac{2}{x-2} \geq 1$

$2x-4 \geq (x-2)^2, x \neq 2$

$2x-4 \geq x^2-4x+4$

$0 \geq x^2-6x+8$

$0 \geq (x-4)(x-2)$



$2 < x \leq 4$

b) $\sin(2\cos^{-1}(\frac{2}{3}))$

Let $\alpha = \cos^{-1}(\frac{2}{3})$
 $\cos \alpha = \frac{2}{3}$



$\sin 2\alpha = 2\sin \alpha \cos \alpha$
 $= 2(\frac{\sqrt{5}}{3})(\frac{2}{3})$

$\sin(2\cos^{-1}(\frac{2}{3})) = \frac{4\sqrt{5}}{9}$

c) $(2x^2 - \frac{1}{3x})^9$ General term

${}^9C_r (2x^2)^r (\frac{1}{3x})^{9-r}$
 $= {}^9C_r 2^r x^{2r} (\frac{1}{3})^{9-r} x^{-9+r}$
 $= {}^9C_r 2^r (-1)^{9-r} (\frac{1}{3})^{9-r} x^{2r-9+r}$
 $6 = 3r-9$
 $r = 5$

$\Rightarrow$ Coefficient of x^6 term
 ${}^9C_5 2^5 (-1)^4 (\frac{1}{3})^4 = \underline{448}$

${}^9C_5 2^5 (-1)^4 (\frac{1}{3})^4 = \frac{448}{9}$

d) 1, 2, 4, 6, 9 > 8000

4 digits, starting with 9 $9___ = 4 \times 3 \times 2 \times 1$
5 digits $5! = 5!$

144 numbers

e) $t = \tan \frac{\theta}{2}$
 $-\pi \leq \theta \leq \pi$

$\sin \theta = \frac{2t}{1+t^2}$ $\cos \theta = \frac{1-t^2}{1+t^2}$

$\frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2} = -\frac{1}{4}$

$4(1+2t-t^2) = -1-t^2$

$4+8t-4t^2 = -t^2-1$

$0 = 3t^2 - 8t - 5$

$t = \frac{8 \pm \sqrt{64+60}}{6}$

$= \frac{4 \pm \sqrt{31}}{3}$

$t = \frac{4+\sqrt{31}}{3}$

$\tan \frac{\theta}{2} = \frac{4+\sqrt{31}}{3}$

$\theta = 2 \tan^{-1}(\frac{4+\sqrt{31}}{3})$
 ≈ 1.27

$t = \frac{4-\sqrt{31}}{3}$

$\tan \frac{\theta}{2} = \frac{4-\sqrt{31}}{3}$

$\theta = 2 \tan^{-1}(\frac{4-\sqrt{31}}{3})$
 ≈ -0.48

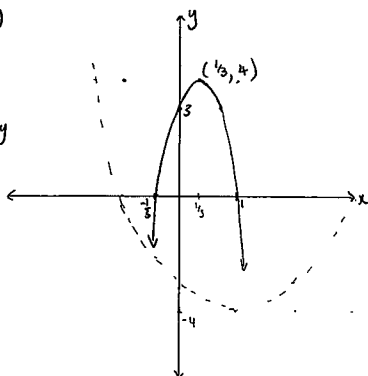
f) $y = (x-1)^2 - 4$

$y = -f(2-3x)$

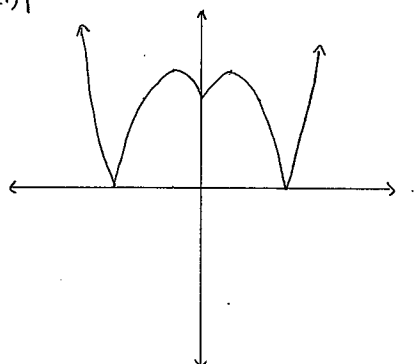


$y = -f(2-3x)$

- $y = f(x)$ (reflect around x)
- $y = f(x+2)$ shift 2 left
- $y = f(-x+2)$ reflect around y
- $y = f(-3x+2)$ dilate x by $\frac{1}{3}$



$y = |f(12x)|$



Q12

Question 12

a) $u^2 = x+1$ $u > 0$ $x=0$ $u^2=1$ $u=1$ $x=3$ $u^2=4$ $u=2$

$\int_0^3 \frac{x+2}{\sqrt{x+1}} dx$

$= \int_1^2 \frac{(u^2+1)2u du}{\sqrt{u^2}}$ 1 mark for substitution

$= \int_1^2 \frac{2u^3+2u}{u} du$

$= \int_1^2 (2u^2+2) du$ 1 mark for taking the integral.

$= [\frac{2u^3}{3} + 2u]_1^2$

$= [\frac{2(8)}{3} + 2(2)] - [\frac{2}{3} + 2]$

$= \frac{16}{3} + 4 - \frac{2}{3} - 2$

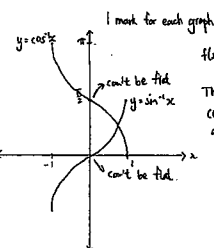
$= 2 + \frac{14}{3}$

$\boxed{\frac{20}{3}}$ 1 mark for final answer.

Any mistake for each step is -1

C.F.E $\rightarrow$ carry forward error.

b) $f(x) = \sin^{-1}x$ $g(x) = \cos^{-1}x$



$f(x) + g(x) = \frac{\pi}{2}$ for $-1 \leq x \leq 1$

This is because $\sin^{-1}x$ and $\cos^{-1}(x)$ are complementary angles.

For significance, must mention they are complementary angles

c) For $n=1$

$$3^{2(1)+2} - 8(1) - 9 = 3^4 - 17 = 81 - 17 = 64 \text{ which is divisible by } 64$$

$\therefore$ True for $n=1$

Step 1 prove true for $n=1$

1 mark

Assume true for $n=k$
i.e. $3^{2k+2} - 8k - 9 = 64m, m \in \mathbb{Z} (*)$

Step 2 assume true for $n=k$
&

Prove true for $n=k+1$

RTP $3^{2(k+1)+2} - 8(k+1) - 9 = 64k, k \in \mathbb{Z}$

Step 3 Prove true for $n=k+1$

LHS = $3^{2k+2+2} - 8k - 8 - 9$
 $= 3^{2k+4} - 8k - 17$
 $= 9 \times 3^{2k+2} - 8k - 17$

1 mark

From $(*)$ $3^{2k+2} = 64m + 8k + 9$

$\therefore$ LHS = $9 \times (64m + 8k + 9) - 8k - 17$
 $= 9 \times 64m + 72k + 81 - 8k - 17$
 $= 9 \times 64m + 64k + 64$
 $= 64(9m + k + 1)$
 which is divisible by 64 since $m, k \in \mathbb{Z}$

Since true for $n=1$ and true for $n=k+1$ it is true for $n=k$
 then by PMI true for all $n \geq 1$

Full conclusion

1 mark

d) 1, 4, 7, 10, ..., 100

Consider pairs which sum to 104
 (100, 4) (97, 7) ... (55, 49)

1 mark for partially explanation

There are 16 such pairs, plus the numbers 1 and 52

(e.g. didn't mention the leftover 1 and 52)

$\therefore$ If choosing 1, 52 and one number from each pair then by the pigeonhole principle the 19th number must be from one of the pairs already chosen.

2 marks for clear explanation using Pigeonhole Principle.

e) $4x^4 + 8x^3 + 8x^2 - 2x - 1 = 0$

Must state the value of a, b, c and d clearly.
 If not, minimum 1 mark.

i) $4x^4 + 8x^3 + 8x^2 - 2x - 1 = 0$
 $(2x^2 + 2x)^2 - (x+1)^2 = 0$

1 mark for getting at least 2 of them right.

$a=2, b=2, c=1, d=1$

2 marks for all values right (note a, b, c, d could also all be negative)

$(2x^2 + 2x - x - 1)(2x^2 + 2x + x + 1) = 0$
 $(2x^2 + x - 1)(2x^2 + 3x + 1) = 0$
 $(2x-1)(x+1)(2x+1)(x+1) = 0$

$\therefore x = -1 \text{ or } x = \frac{1}{2} \text{ or } x = -\frac{1}{2}$

1 mark for at least 1 right.

2 marks for all 3 values right (no extra)

13(a) i) $f(x) = x \cos^{-1} x - \sqrt{1-x^2}$
 $f'(x) = \left[\cos^{-1} x - \frac{x}{\sqrt{1-x^2}} \right] + \frac{x}{\sqrt{1-x^2}}$
 $= \cos^{-1} x$

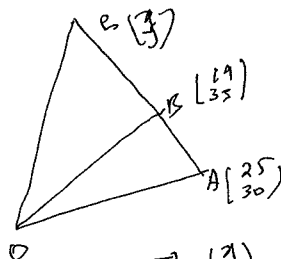
ii) $\int_0^1 \cos^{-1} x \, dx = \left[x \cos^{-1} x - \sqrt{1-x^2} \right]_0^1$
 $= 0 - (0 - \sqrt{1})$
 $= 1$

(b) ii) $T = A + Ce^{kt}$
 $t=0, A=15^\circ, T=25^\circ$
 $\Rightarrow 25 = 15 + C \Rightarrow C=10$

$t = 30 \text{ min}$
 $T = 21^\circ$
 $21 = 15 + 10e$
 $k = \frac{\ln 0.6}{30}$
 ≈ -0.017
 $t = \frac{1}{k} \ln(0.6)$
 $k = 2 \ln(0.6)$
 ≈ -0.017

iii) $T = 16^\circ$
 $16 = 15 + 10e^{kt}$
 $\Rightarrow t = \frac{1}{k} \ln(0.1)$
 $k = \frac{\ln(0.6)}{30} \Rightarrow t = \frac{\ln(0.1)}{2 \ln(0.6)} = \frac{\ln(0.1)}{2 \ln(0.6)} \approx 135.21 \dots$
 $k = 2 \ln(0.6) \Rightarrow t = \frac{\ln(0.1)}{2 \ln(0.6)} = \frac{\ln(0.1)}{2 \ln(0.6)} \approx 2.25378 \dots$
 $\approx 2 \text{ h } 15 \text{ min}$

13 (c) ii)



Position vector of $C = \vec{OC} = \begin{pmatrix} 11 \\ 9 \end{pmatrix}$

$\begin{pmatrix} x \\ y \end{pmatrix} = 5 \begin{pmatrix} 25 \\ 30 \end{pmatrix} + \vec{OA}$
 $= 5 \begin{pmatrix} -6 \\ 3 \end{pmatrix} + \begin{pmatrix} 25 \\ 30 \end{pmatrix}$
 $= \begin{pmatrix} -5 \\ 45 \end{pmatrix}$

ii) $\cos \angle AOC = \frac{\vec{OA} \cdot \vec{OC}}{|\vec{OA}| |\vec{OC}|}$

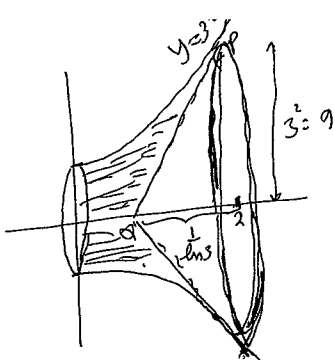
$\vec{OA} \cdot \vec{OC} = \begin{pmatrix} 25 \\ 30 \end{pmatrix} \cdot \begin{pmatrix} -5 \\ 45 \end{pmatrix} = -1225$
 $|\vec{OA}| = \sqrt{25^2 + 30^2} = 5\sqrt{61}$
 $|\vec{OC}| = \sqrt{(-5)^2 + 45^2} = 5\sqrt{82}$

$\therefore \cos \angle AOC = \frac{-1225}{25\sqrt{61} \cdot 5\sqrt{82}}$

$\approx 0.6928244 \dots$

$\Rightarrow \angle AOC = 46^\circ 9' \text{ (nearest min)}$

13(c)



Vol. of the required solid = (Vol. of solid generated by rotating $y = 3 - x^2$ about the x-axis between $x = -2$ and $x = 2$)

(Vol. of solid generated by rotating $y = 3 - x^2$ about the x-axis between $x = -2$ and $x = 2$)

$$\begin{aligned} V &= \pi \int_{-2}^2 (3 - x^2)^2 dx - (\text{Vol. of cone of radius 3 and height } \frac{1}{2}) \\ &= \pi \left[\frac{3^3 x}{3} - \frac{2 \cdot 3^2 x^2}{2} + \frac{x^3}{3} \right]_{-2}^2 - \frac{1}{3} \pi (3^2) \left(\frac{1}{2} \right) \\ &= \pi \left[\frac{81}{3} - \frac{1}{2} \right] - \frac{27\pi}{6} \\ &= \frac{13\pi}{6} \text{ units}^3 \end{aligned}$$

iii)

$$\begin{aligned} &\int_0^k \frac{1 - \frac{e^x - e^{-x}}{e^x + e^{-x}}}{\frac{e^x + e^{-x}}{e^x + e^{-x}}} dx \\ &= [x - \ln(e^x + e^{-x})]_0^k \\ &= k - \ln(e^k + e^{-k}) - 0 + \ln 2 \\ &= \ln 2 + \ln \left(\frac{2e^k}{e^k + e^{-k}} \right) \\ &= \ln \left(\frac{2e^k}{e^k + e^{-k}} \right) \end{aligned}$$

iv) from iii) $A = \ln \left(\frac{2e^k}{e^k + e^{-k}} \right)$

$$\lim_{k \rightarrow \infty} \ln \left(\frac{2e^k}{e^k + e^{-k}} \right) = \ln 2 + \ln \left(\frac{e^k}{e^k + e^{-k}} \right)$$

As $k \rightarrow \infty$, $e^{-k} \rightarrow 0$

$$\therefore \ln \left(\frac{2e^k}{e^k + e^{-k}} \right) \text{ approaches } \ln \left(\frac{2e^k}{e^k} \right) = \ln 2$$

Alternatively, consider $\ln \left(\frac{2e^k}{e^k + e^{-k}} \right) = \ln 2 + \ln \left(\frac{e^k}{e^k + e^{-k}} \right)$

Since e^k, e^{-k} are both > 0

$$e^k < e^k + e^{-k}$$

$$\therefore \frac{e^k}{e^k + e^{-k}} < 1$$

$$\therefore \ln \left(\frac{e^k}{e^k + e^{-k}} \right) < 0$$

Q14

Tuesday, 6 August 2014 9:50 AM

Question 14

$$\sin x (6 \sin x + 4 \cos x - 3) = 0$$

$$\sin x = 0 \quad \text{or} \quad 6 \sin x + 4 \cos x = 3$$

$$x = 0, \pi$$

$$\sqrt{5} \sin \left(x + \tan^{-1} \left(\frac{3}{4} \right) \right) = 3$$

$$\sin \left(x + \tan^{-1} \left(\frac{3}{4} \right) \right) = \frac{3}{\sqrt{5}}$$

$$x + \tan^{-1} \left(\frac{3}{4} \right) = \sin^{-1} \left(\frac{3}{\sqrt{5}} \right), \pi - \sin^{-1} \left(\frac{3}{\sqrt{5}} \right)$$

$$x = 0.5880 \dots, \pi - 0.42906 \dots$$

$$x = -0.158 \dots, 2.124 \dots$$

$$x = 2.12 \quad 0 \leq x \leq \pi$$

$$\therefore x = 0, 2.12, \pi$$

-could also use t-method or similar. Answers in radians

b) i) $f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

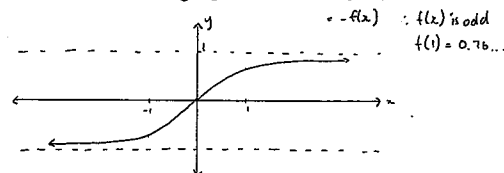
$$f'(x) = \frac{(e^x + e^{-x})^2 - (e^x - e^{-x})^2}{(e^x + e^{-x})^4}$$

$$0 = 1 - \frac{(e^x - e^{-x})^2}{(e^x + e^{-x})^2}$$

$$1 = \frac{e^{2x} - 2 + e^{-2x}}{e^{2x} + 2 + e^{-2x}}$$

$$e^{2x} + 2 + e^{-2x} = e^{2x} - 2 + e^{-2x} \quad \text{Since } 2 \neq -2, \text{ no stationary points}$$

ii) $y = \pm 1 \quad f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad f(-x) = \frac{e^{-x} - e^x}{e^{-x} + e^x} = -f(x)$



i)



$$\begin{aligned} y &= -10 \quad x = 0 \\ y &= -10t + C \quad x = C \\ \text{At } t = 0, 80 \cos \alpha = C &\quad x = 80 \cos \alpha \\ \therefore y &= -10t + 80 \sin \alpha \quad x = 80 \sin \alpha + C \\ y &= -5t^2 + 80t \sin \alpha + C &\quad x = 80t \cos \alpha + C \\ \text{At } t = 0, y = 0 &\quad C = 0 \\ &\quad x = 0 \quad x = 80t \cos \alpha \end{aligned}$$

$$\therefore y = -5t^2 + 80t \sin \alpha \quad x = 80t \cos \alpha$$

$$s(t) = 80t \cos \alpha \hat{i} + (80t \sin \alpha - 5t^2) \hat{j}$$

ii) $s_2(t)$ for particle 2

$$s_2(t) = Vt \cos \beta \hat{i} + (Vt \sin \beta - 5t^2) \hat{j}$$

Particles collide when $s_1(t)$ and $s_2(t)$

$$\therefore V \cos \beta = 80 \times 3 \cos \alpha \quad \text{and} \quad V \sin \beta - 5 = 80 \times 3 \sin \alpha - 45$$



$$V \cos \beta = 240 \left(\frac{12}{13} \right)$$

$$= \frac{2880}{13}$$

$$V \sin \beta = 240 \times \frac{5}{13} - 40$$

$$= \frac{680}{13}$$

$$\therefore \tan \beta = \frac{680}{2880}$$

$$= \frac{17}{72}$$

iii) From i)

$$V \cos \beta = \frac{2880}{13}$$



$$V \left(\frac{72}{72.5} \right) = \frac{2880}{13}$$

$$V = \frac{2880}{13} \times \frac{72.5}{72}$$

$$= \frac{40 \sqrt{5473}}{13}$$

$$\approx 227.43 \text{ m/s (2dp)}$$